

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12405183
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Hsiao; Chieh-Jen

Adjustment device for torque wrench

Abstract

An adjustment device for adjusting a torque of a torque wrench includes a base unit, a driving unit, and a control unit. The torque wrench includes a shank body extending along an axis, an adjusting unit threadedly engaging the shank body, and a resilient member abutting against the adjusting unit. The base unit is for mounting of the torque wrench. The driving unit includes a coupling member removably connected to the adjusting unit, and a driving motor. The control unit controls the driving motor to drive rotation of the coupling member about the axis for driving rotation of the adjusting unit to thereby move the adjusting unit along the axis to vary a preload force of the resilient member and thus a torque of the torque wrench and to move the adjusting unit to a position where the torque of the torque wrench corresponds to a predetermined torque value.

Inventors:	Hsiao; Chieh-Jen (Taichung, TW)
Applicant:	Hsiao; Fang-Wei (Taichung, TW)
Family ID:	88585748
Assignee:	Hsiao; Fang-Wei (Taichung, TW)
Appl. No.:	18/449424
Filed:	August 14, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240068899 A1	Feb. 29, 2024

Foreign Application Priority Data

TW	111131882	Aug. 24, 2022
----	-----------	---------------

Publication Classification

Int. Cl.: G01L25/00 (20060101); B25B23/142 (20060101)

U.S. Cl.:

CPC G01L25/003 (20130101); B25B23/1427 (20130101);

Field of Classification Search

CPC: G01L (25/003); B25B (23/1427)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2009/0222222	12/2008	Lucke	702/41	G01L 25/003
2024/0286254	12/2023	Chen	N/A	B25B 21/026

Primary Examiner: Sinha; Tarun

Attorney, Agent or Firm: Burris Law, PLLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Taiwanese Invention Patent Application No. 111131882, filed on Aug. 24, 2022.

FIELD

(2) The disclosure relates to an adjustment device, and more particularly to an adjustment device for a torque wrench.

BACKGROUND

(3) A conventional adjustment device is provided for calibrating a torque of a torque wrench. The torque wrench includes a shank body, a torsion head mounted on one end of the shank body, a handle threadedly engaging the other end of the shank body, and a spring disposed in the shank body, and clamped between and abutting against the torsion head and the handle.

(4) The conventional adjustment device includes a base for mounting of the torque wrench, two limiting members extending upwardly from the base and for clamping the shank body, a torque device mounted on the base, a display, and a computer electrically connected to the torque device and the display. The torque device includes a rotating member that is rotatable and that co-rotatably engages the torsion head of the torque wrench, a motor that is electrically connected to the computer to be controlled thereby and that is operable for driving rotation of the rotating member, and a torque sensor that measures the torque exerted by the rotating member and that is electrically connected to the computer to transmit a measuring result thereto.

(5) When it is desired to calibrate the torque of the torque wrench, the handle is first rotated so the handle is moved relative to the shank body to compress or to release the spring, and thus the torque of the torque wrench is adjusted to correspond to a predetermined torque value. Then, the torsion head of the torque wrench is brought to engage the rotating member, the shank body is placed to be clamped between the limiting members, and the computer is operated to control the motor to drive rotation of the rotating member and thus the torque head. The torque sensor measures a peak torque

value exerted by the rotating member when the torque exerted by the rotating member is greater than an actual torque value of the torque wrench, which is provided by the spring. When a difference between the peak torque value and the predetermined torque value is greater than a tolerance, the spring is to be replaced by another spring having an elasticity coefficient different from that of the original spring according to magnitude of the difference. Hereafter, the abovementioned operations are repeated until the difference is smaller than the tolerance. For example, in a case where a torque wrench is supposed to have a predetermined torque value equal to 40 Newton metre (Nm), and the peak torque value obtained from the torque sensor is 41 Nm, which represents that the actual torque value of the torque wrench is greater than the predetermined torque value. Therefore, the original spring is to be replaced by another spring with an elastic coefficient smaller than that of the original spring, until the peak torque value is substantially equal to the predetermined torque value, i.e., 40 Nm.

(6) However, the abovementioned operation conducted by the conventional adjustment device is a trial-and-error method. It is troublesome to manually rotate the handle, to measure the peak torque value, and to replace the spring. Thus, calibration performed by the conventional adjustment device is not only time-consuming but also labor-intensive.

SUMMARY

(7) Therefore, an object of the disclosure is to provide an adjustment device and that can alleviate at least one of the drawbacks of the prior art.

(8) According to an aspect of the disclosure, an adjustment device for adjusting a torque of a torque wrench is provided. The torque wrench includes a shank body that extends along an axis and that defines an accommodating space therein, a torsion member and an adjusting unit that are disposed at two opposite ends of the shank body along the axis, and a resilient member that is disposed in the accommodating space and clamped between the torsion member and the adjusting unit. The adjusting unit threadedly engages the shank body, and is rotatable about the axis so as to be movable relative to the shank body along the axis to vary a preload force of the resilient member and thus a torque of the torque wrench. The adjustment device includes a base unit, a driving unit, and a control unit. The base unit includes a frame seat adapted for mounting of the torque wrench thereon, and a positioning member disposed on the frame seat and adapted for positioning the shank body. The driving unit includes a coupling member adapted to be removably connected to the adjusting unit, and a driving motor mounted movably to the frame seat, and operable for driving the coupling member to rotate about the axis for driving the adjusting unit to rotate about the axis. The control unit is electrically connected to the driving motor, and is configured to control the driving motor to drive rotation of the coupling member about the axis for driving rotation of the adjusting unit to thereby move the adjusting unit along the axis to a position where the torque of the torque wrench corresponds to a predetermined torque value.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Other features and advantages of the disclosure will become apparent in the following detailed description of the embodiment(s) with reference to the accompanying drawings. It is noted that various features may not be drawn to scale.

(2) FIG. 1 is a block diagram of an adjustment device of a first embodiment according to the present disclosure.

(3) FIG. 2 is a schematic top view of the first embodiment.

(4) FIG. 3 is a sectional view of a torque wrench to be adjusted by the first embodiment.

(5) FIG. 4 is a schematic view illustrating the torque wrench being mounted to a base unit of the adjustment device of the first embodiment.

(6) FIG. 5 is a flowchart illustrating a method for adjusting a torque of the torque wrench conducted by the first embodiment.

(7) FIG. 6 is a schematic view illustrating the torque wrench being mounted on an adjustment device according to a second embodiment of the present disclosure.

DETAILED DESCRIPTION

(8) It should be noted herein that for clarity of description, spatially relative terms such as “top,” “bottom,” “upper,” “lower,” “on,” “above,” “over,” “downwardly,” “upwardly” and the like may be used throughout the disclosure while making reference to the features as illustrated in the drawings. The features may be oriented differently (e.g., rotated 90 degrees or at other orientations) and the spatially relative terms used herein may be interpreted accordingly.

(9) Referring to FIGS. 1 to 3, an adjustment device of a first embodiment according to the present disclosure is for adjusting a torque of a torque wrench 6. The torque wrench 6 includes a shank body 61, a resilient member 62, a torsion member 63, an adjusting unit 64, and an articulate rod module 65. The shank body 61 is hollow, extends along an axis (L), and defines an accommodating space 611 therein. The resilient member 62 is disposed in the accommodating space 611 and is clamped between the torsion member 63 and the adjusting unit 64. The adjusting unit 64 threadedly engages the shank body 61, and is rotatable about the axis (L) so as to be movable relative to the shank body 61 along the axis (L) to vary a preload force of the resilient member 62 and thus a torque of the torque wrench 6.

(10) The adjustment device includes a base unit 1, a driving unit 2, a force applying unit 3, a calibration unit 4, and a control unit 5.

(11) The base unit 1 includes a frame seat 11, two sliding rail sets 12, 12' disposed on the frame seat 11, and a positioning member 13 including two clamping portions 131 that are disposed on the frame seat 11. The frame seat 11 has a front side 101, a rear side 102 opposite to the front side 101, a left side 103 interconnecting the front side 101 and the rear side 102, and a right side 104 opposite to the left side 103 and interconnecting the front side 101 and the rear side 102.

(12) In this embodiment, the sliding rail sets 12, 12' are mounted respectively adjacent to the rear side 102 and the left side 103 of the frame seat 11.

(13) The driving unit 2 includes a driving motor 21 and a coupling member 22. The coupling member 22 includes a sleeve 221, and is adapted to be removably connected to the adjusting unit 64. The driving motor 21 is mounted movably to the frame seat 11, is disposed between one of the sliding rail sets 12 adjacent to the rear side 102 of the frame seat 11, and is operable for driving the coupling member 22 to rotate about the axis (L) for driving the adjusting unit 64 to rotate about the axis (L).

(14) The force applying unit 3 includes a rotating member 31, a torque-exerting motor 32, and a measuring module 33. The rotating member 31 is rotatably mounted adjacent to the front side 101 on the frame seat 11, is disposed opposite to the driving motor 21, and is adapted for engaging the torsion member 63 and exerting a torque on the torsion member 63 to drive rotation of the torsion member 63. In this embodiment, the rotating member 31 has a recess 311 that is adapted for engaging the torsion member 63. The torque-exerting motor 32 is mounted on the frame seat 11 for driving rotation of the rotating member 31. The measuring module 33 is mounted to the rotating member 31, and is configured to measure a peak torque value exerted by the rotating member 31 when the value of the torque exerted by the rotating member 31 is greater than an actual torque value of the torque wrench 6. It should be noted that in this embodiment, the torque-exerting motor 32 and the measuring module 33 are mounted under the frame seat 11 and are depicted by dashed lines in the drawings.

(15) The calibration unit 4 includes a calibration member 41, a calibration motor 42, and a restoring member 43. The calibration member 41 is disposed between the other one of the sliding set 12' on the frame seat 11, and is adapted to extend into a calibration hole 612 that is formed in the shank body 61 and that is in spatial communication with the accommodating space 611. The calibration

member **41** is further adapted to be connected to the articulate rod module **65** of the torque wrench **6**. The calibrating motor **42** is disposed on the frame seat **11**, is electrically connected to the control unit **5**, and is operable for driving rotation of the calibration member **41**. The restoring member **43** is connected between the calibration member **41** and the calibration motor **42** for resiliently biasing the calibrating member **41** away from the calibrating motor **42**. In this embodiment, the restoring member **43** is made of an elastic material, and biases the calibration member **41** away from the calibration motor **42** so as to be exposed outwardly of the calibration motor **42**. In this way, the calibration member **41** is movable relative to the calibration motor **42** in a resilient manner.

(16) In this embodiment, the calibration member **41** has one end that is opposite to the calibration motor **42** and that is hexagonal. In other variations of the present disclosure, the end of the calibration member **41** may be a flat head, a cross head or have other configurations, and the present disclosure is not limited hereto.

(17) The control unit **5** includes a display module **51**, an input module **52**, a storage module **53**, and a control module **54**. The storage module **53** of the control unit **5** stores a plurality of setting torque values. It should be noted that, prior to adjusting a torque of the torque wrench **6**, a plurality of to-be-measured torque values may be inputted by a user via the input module **52** so as to be stored in the storage module **53** and respectively serve as the setting torque values. The control module **54** is electrically connected to the measuring module **33**, the torque-exerting motor **32**, the driving motor **21**, the display module **51**, the input module **52**, and the storage module **53**. The control module **54** is further configured to receive the peak torque value from the measuring module **33** and transmit the same to be stored in the storage module **53**. The control module **54** is configured to control operations of the driving motor **21**, the torque-exerting motor **32**, and the calibration motor **42**. Specifically, the control module **54** drives rotation of the driving motor **21** to drive rotation of the coupling member **22** about the axis (L) for driving rotation of the adjusting unit **64** to thereby move the adjusting unit **64** along the axis (L) to a position where the torque of the torque wrench **6** corresponds to a predetermined torque value. In this embodiment, the display module **51** is a screen, and the input module **52** is a keyboard, and other display modules and input modules may be used in other embodiments of the present disclosure. It should be noted that the control module **54** includes a microcontroller or a controller such as, but not limited to, a single core processor, a multi-core processor, a dual-core mobile processor, a microprocessor, a microcontroller, a digital signal processor (DSP), a field-programmable gate array (FPGA), an application specific integrated circuit (ASIC), a radio-frequency integrated circuit (RFIC), etc.

(18) As shown in FIG. 3, the resilient member **62** has a head end **621**, and a tail end **622** opposite to the head end **621** along the axis (L). In this embodiment, the resilient member **62** is a spring.

(19) The torsion member **63** and the adjusting unit **64** are disposed at two opposite ends of the shank body **61** along the axis (L). The torsion member **63** has a head portion **631**, and a body portion **632** connected to the head portion **631**, extending into the accommodating space **611**, and connected pivotably to the shank body **61**. The adjusting unit **64** includes an abutment portion **641**, a polygonal rod **642**, and a handle portion **643**. The abutment portion **641** threadably engages the shank body **61**, abuts against the tail end **622** of the resilient member **62**, and is co-rotatably connected to and abuts against the polygonal rod **642** at an end thereof opposite to the resilient member **62**. The handle portion **643** is sleeved movably on the shank body **61**, and is detachably connected to the abutment portion **641**. The polygonal rod **642** extends into the tail end **622** of the resilient member **62**, and is driven by the driving motor **21** to drive rotation of the abutment portion **641** to move the abutment portion **641** along the axis (L).

(20) The head end **621** of the resilient member **62** is connected to and abuts against the articulate rod module **65**, and the tail end **622** of the resilient member **62** abuts against the abutment portion **641** of the adjusting unit **64**. In this way, the resilient member **62** is compressed and released during rotation of the adjusting unit **64** about the axis (L) and thus movement of the adjusting unit **64** along the axis (L) relative to the shank body **61**.

(21) In this embodiment, a length of the articulate rod module **65** along the axis (L) is adjustable so as to fine tune the preload force of the resilient member **62** and thus the torque of the torque wrench **6**. It should be noted that, in other embodiments of the present disclosure, the articulate rod module **65** may be replaced by a slidable structure (not shown) that is movable relative to the shank body **61** along the axis (L) and that directly engages the resilient member **62**, such that the resilient member **62** is resiliently clamped between the torsion member **63** and the adjusting unit **64**, and the preload force thereof may also be adjusted when the slidable structure moves along the axis (L). In this embodiment, the articulate rod module **65** is disposed in the accommodating space **611**, is pivotably connected to the torsion member **63**, and includes a slidable block **651**, a first link rod **652**, a second link rod **653**, and a threaded rod **654**. The slidable block **651** abuts against the head end **621** of the resilient member **62**. The first link rod **652** has a first end portion **655** abutting against the body portion **632** of the torsion member **63**, a second end portion **656** opposite to the first end portion **655**, a pivot connecting portion **657** located between the first end portion **655** and the second end portion **656** and pivotably connected to the shank body **61** via a pin, and a threaded hole **658** formed between the pivot connecting portion **657** and the second end portion **656** and extending transverse to the axis (L). Two opposite ends of the second link rod **653** are pivotably and respectively connected to the second end portion **656** and the slidable block **651**. The threaded rod **654** extends through and threadably engages the threaded hole **658**, and has first and second ends opposite in a direction transverse to the axis (L). The first end of the threaded rod **654** is disposed adjacent to the shank body **61**, and the second end of the threaded rod **654** is formed with an engaging hole **659** facing the calibration hole **612**. When the slidable block **651** is biased by the resilient member **62** to push the second link rod **653** along the axis (L) via one of the ends of the second link rod **653** that is pivotably connected thereto, the other one of the ends of the second link rod **653** that is pivotably connected to the second end portion **656** of the first link rod **652** moves the first end of the threaded rod **654** to press against the shank body **61**. An arrow shown in FIG. 3 depicts a direction of torque being exerted on the first link rod **652**.

(22) The engaging hole **659** is an internal hexagonal hole being engaged with and co-rotatable with the calibration member **41**. It should be noted that the shape of the engaging hole **659** may be modified as long as the engaging hole **659** may be engaged with and co-rotatable with the calibration member **41**, and the present disclosure is not limited to the configuration of the engaging hole **659**.

(23) When the calibration rod **41** is driven by the calibration motor **42** to rotate, the threaded rod **654** is co-rotatable therewith through engagement between the engaging hole **659** and the calibration rod **41**. Specifically, the calibration rod **41** is rotatable in a first rotational direction for expanding the articulate rod module **65** to compress the resilient member **62**, and is rotatable in a second rotational direction opposite to the first rotational direction for shrinking the articulate rod module **65** to release the resilient member **62**, thereby varying the preload force of the resilient member **62**. In this embodiment, when the threaded rod **654** is rotated by the calibration rod **41** in the first rotational direction, the threaded rod **654** abuts against the shank body **61** and the first link rod **652** pivots relative to the second link rod **653**, and is moved in a direction depicted by the arrow shown in FIG. 4, such that an angle between the first link rod **652** and the second link rod **653** is increased to push the slidable block **651** toward the resilient member **62** and thus the articulate rod module **65** expands to compress the resilient member **62** along the axis (L). On the other hand, when the threaded rod **654** is rotated by the calibration rod **41** in the second rotational direction, the threaded rod **654** still abuts against the shank body **61** and the first link rod **652** pivots relative to the second link rod **654**, and is moved in a direction depicted by an arrow shown in FIG. 3, such that an angle between the first link rod **652** and the second link rod **653** is decreased and the slidable block **651** is moved by the second link rod **653** away from resilient member **62** and thus the articulate rod module **65** is shrunk to release the resilient member **62**. In this way, the preload force of the resilient member **62** is adjusted and thus the torque of the torque wrench **6** may be

finely tuned by rotation of the calibration member **41** via the threaded rod **654** without rotating the handle **643**. In this embodiment, the first rotational direction is a clockwise direction of the threaded rod **654** relative to the first link rod **652** in FIG. **3**, while the second rotational direction is a counterclockwise direction of the threaded rod **654** relative to the first link rod **652** in FIG. **3**.

(24) In this embodiment, the output torque value of the torque wrench **6** ranges from 20 Nm to 100 Nm, but is not limited thereto.

(25) As shown in FIGS. **3** and **4**, steps of mounting the torque wrench **6** on the frame seat **11** of the base unit **1** are described in the following. First, the handle **643** of the adjusting unit **64** is removed from the shank body **61**, and the remaining portion of the torque wrench **6** is disposed on the frame seat **11** so that the clamping portions **131** of the positioning member **13** clamp the shank body **61** therebetween. In this embodiment, the clamping portions **131** are adapted to clamp diametrically opposite sides of the shank body **61**. The recess **311** of the rotating member **31** of the force applying unit **3** is adapted for engaging the head portion **631** of the torsion member **63** and exerting a torque on the torsion member **63** to drive rotation of the torsion member **63**. The sleeve **221** of the coupling member **22** of the driving unit **2** is adapted to be sleeved on and drive the polygonal rod **642** to rotate, thereby driving the abutment portion **641** of the adjusting unit **64** to move along the axis (L). In this embodiment, the polygonal rod **642** is a hexagonal rod, and the sleeve **221** is internal hexagonal and is removably sleeved on the polygonal rod **642**. Then, the calibration motor **42** is controlled by the control module **54** to drive the calibration member **41** to extend into the calibration hole **612** and engage the engaging hole **659** formed in the threaded rod **654**. When the rotating member **31** exerts a torque on the head portion **631** of the torsion member **63** to drive rotation thereof, a direction of the torque exerted on the first link rod **652** is opposite to the direction depicted by the arrow shown in FIG. **4**. It should be noted that, in a case where the torque exerted by the rotating member **31** is smaller than the actual torque value of the torque wrench **6**, the first link rod **652** will not be rotated.

(26) Referring to FIG. **5**, a method for adjusting a torque of the torque wrench **6** to be conducted by the adjustment device of the first embodiment of the present disclosure is shown and includes steps **S1** to **S9**.

(27) In step **S1**, the control module **54** selects a minimum one of the setting torque values stored in the storage module **53** to be the predetermined torque value.

(28) In step **S2**, the control module **54** controls the driving motor **21** to drive rotation of the coupling member **22** about the axis (L) for driving the polygonal rod **642** of the adjusting unit **64** to rotate about the axis (L), so the adjusting unit **64** is moved relative to the shank body **61** along the axis (L) to a position where the torque of the torque wrench **6** corresponds to the predetermined torque value.

(29) In step **S3**, the control module **54** controls the torque-exerting motor **32** to drive rotation of the rotating member **31** to exert a torque on the head portion **631** of the torsion member **63** and to drive rotation of the head portion **631**, and the measuring module **33** measures a peak torque value exerted by the rotating member **31** when the value of the torque exerted by the rotating member **31** is larger than an actual torque of the torque wrench **6**. At this time, the control module **54** receives the peak torque value from the measuring module **33** and transmits the peak torque value to be stored in the storage module **53**, and then the control module **54** controls the torque-exerting motor **32** to stop driving the rotating member **31** to thereby release the torsion member **63**.

(30) In step **S4**, the control module **54** determines whether each of the setting torque values stored in the storage module **53** is selected as the predetermined torque value. When the determination is affirmative, the flow of the method proceeds to step **S6**, otherwise the flow goes to step **S5**.

(31) In step **S5**, the control module **54** selects the minimum one of the setting torque values that has not been selected as the predetermined torque value and the flow of the method goes back to step **S2**.

(32) When the control module **54** determines that each of the setting torque values stored in the

storage module **53** is selected as the predetermined torque value, step **S6** is performed. In step **S6**, the control module **54** generates a linear equation according to the predetermined torque value and the peak torque value obtained from step **S3** corresponding to each of the setting torque values stored in the storage module **53** and then the flow of the method goes to step **S7**. For example, in this embodiment, the linear equation is expressed as $Y=aX+b$, where X and Y respectively represent the peak torque value and the predetermined torque value corresponding to a selected one of the setting torque values, and a and b represents an upper limit and a lower limit of a calibration tolerance.

(33) In step **S7**, the control module **54** determines whether the linear equation satisfies a calibration tolerance standard, and the flow of method proceeds to step **S9** when affirmative, otherwise step **S8** is performed. For example, the control module **54** determines that the linear equation satisfies the calibration tolerance standard when a is greater than 0, and b is greater than 0 and smaller than 1. In a case where a is equal to 1 and b is equal to 0, X is equal to Y . It should be noted that a range of the calibration tolerance may be modified as required, and the main feature of the present disclosure does not reside in the determination made in step **S7**, further details of the same are omitted for the sake of brevity.

(34) In step **S8**, the control module **54** controls the calibration motor **42** to drive rotation of the calibration member **41** and thus the threaded rod **654** according to the linear equation to fine tune the preload force of the resilient member **62** and thus the torque of the torque wrench **6**, then the flow of the method goes back to step **S1**. It should be noted that how the control module **54** controls the calibration motor **42** according to the linear equation is well known in the pertinent art and is not the salient feature of the present disclosure. Therefore, further details elaborating the same are omitted for the sake of brevity.

(35) In step **S9**, the control module **54** controls the display module **51** to display the linear equation.

(36) It should be noted that in one embodiment of the present disclosure, the control module **54** controls the display module **51**, in steps **S1** to **S6**, to display the information of the predetermined torque value corresponding to the selected one of the setting torque values, the peak torque value, and the linear equation for the user to refer to or record in real time.

(37) In this embodiment, the number of the to-be-measured torque values is three and the to-be-measured torque values are respectively, e.g., 20 Nm, 60 Nm, and 100 Nm. The storage module **53** stores a conversion equation related to the number of rotation of the abutment portion **641** corresponding to a variation in the torque of the torque wrench **6**. Thus, the control unit **5** controls the driving motor **21** to rotate the abutment portion **641** for a certain number of turns according to the conversion equation so as to move the abutment portion **641** to a position where the torque of the torque wrench **6** corresponds to the predetermined torque value, i.e., the selected one of the setting torque values. In other embodiments of the present disclosure, when another torque wrench having a torque different from that of the torque wrench **6** is to be adjusted, another conversion equation related to the number of rotation corresponding to a variation in the torque of another torque wrench is stored in the storage module **53**. It should be noted that, the number of the to-be-measured torque values may be modified to be two or more than four and the present disclosure is not limited thereto.

(38) Referring to FIG. **6**, the adjustment device of a second embodiment according to the present disclosure is similar to the first embodiment, and the differences between the first embodiment and the second embodiment reside in the following. In the second embodiment, the coupling member **22** includes a fastening portion **222** adapted to be fixedly connected to and drive the handle portion **643** of the adjusting unit **64** to rotate, thereby driving the abutment portion **641** of the adjusting unit **64**, which is co-rotatably connected to the handle portion **643**, which threadedly engages the shank body **61**, and which abuts against the resilient member **62**, to move along the axis (L). In this embodiment, it is not required to remove the handle **643** when mounting the torque wrench **6** on the frame seat **11** of the base unit **1**.

(39) In summary, in the adjustment device of the present disclosure, by virtue of the driving motor **21** that drives rotation and thus movement of the adjusting unit **64** along the axis (L), a process of manually rotating the handle **643** to adjust the torque of the torque wrench **6** may be omitted. Furthermore, the calibration unit **4** is controlled by the control module **54** to fine tune the preload force of the resilient member **62** and thus the torque wrench **6**, so calibration of the torque wrench **6** may be completed in a relatively simple manner. In addition, by virtue of the restoring member **43** that biases the calibrating member **41** away from the calibrating motor **42**, the calibrating member **41** engages the engaging hole **659** at all time, and removal of the calibration member **41** from the engaging hole **659** during rotation of the rotating member **31** that drives rotation of the torsion member **63** to drive pivot movement of the first link rod **652** may be prevented.

(40) In the description above, for the purposes of explanation, numerous specific details have been set forth in order to provide a thorough understanding of the embodiment(s). It will be apparent, however, to one skilled in the art, that one or more other embodiments may be practiced without some of these specific details. It should also be appreciated that reference throughout this specification to “one embodiment,” “an embodiment,” an embodiment with an indication of an ordinal number and so forth means that a particular feature, structure, or characteristic may be included in the practice of the disclosure. It should be further appreciated that in the description, various features are sometimes grouped together in a single embodiment, figure, or description thereof for the purpose of streamlining the disclosure and aiding in the understanding of various inventive aspects; such does not mean that every one of these features needs to be practiced with the presence of all the other features. In other words, in any described embodiment, when implementation of one or more features or specific details does not affect implementation of another one or more features or specific details, said one or more features may be singled out and practiced alone without said another one or more features or specific details. It should be further noted that one or more features or specific details from one embodiment may be practiced together with one or more features or specific details from another embodiment, where appropriate, in the practice of the disclosure.

(41) While the disclosure has been described in connection with what is(are) considered the exemplary embodiment(s), it is understood that this disclosure is not limited to the disclosed embodiment(s) but is intended to cover various arrangements included within the spirit and scope of the broadest interpretation so as to encompass all such modifications and equivalent arrangements.

Claims

1. An adjustment device for adjusting a torque of a torque wrench, the torque wrench including a shank body that extends along an axis and that defines an accommodating space therein, a torsion member and an adjusting unit that are disposed at two opposite ends of the shank body along the axis, and a resilient member that is disposed in the accommodating space and clamped between the torsion member and the adjusting unit, the adjusting unit threadably engaging the shank body and rotatable about the axis so as to be movable relative to the shank body along the axis to vary a preload force of the resilient member and thus a torque of the torque wrench, said adjustment device comprising: a base unit that includes a frame seat adapted for mounting of the torque wrench thereon, and a positioning member disposed on said frame seat and adapted for positioning the shank body; a driving unit that includes a coupling member adapted to be removably connected to the adjusting unit, and a driving motor mounted movably to said frame seat, and operable for driving said coupling member to rotate about the axis for driving the adjusting unit to rotate about the axis; and a control unit that is electrically connected to said driving motor, and that is configured to control said driving motor to drive rotation of said coupling member about the axis for driving rotation of the adjusting unit to thereby move the adjusting unit along the axis to a

position where the torque of the torque wrench corresponds to a predetermined torque value.

2. The adjustment device as claimed in claim 1, further comprising a force applying unit that includes: a rotating member rotatably mounted on said frame seat, and adapted for engaging the torsion member and exerting a torque on the torsion member to drive rotation of the torsion member; a torque-exerting motor mounted on said frame seat for driving rotation of said rotating member; and a measuring module mounted to said rotating member and configured to measure a peak torque value exerted by said rotating member when the value of the torque exerted by said rotating member is greater than an actual torque value of the torque wrench.

3. The adjustment device as claimed in claim 2, wherein said control unit includes: a control module electrically connected to said measuring module, said torque-exerting motor, and said driving motor, and configured to control operations of said driving motor and said torque-exerting motor; and a storage module electrically connected to said control module and storing a plurality of setting torque values including the predetermined torque value, said control module being configured to receive the peak torque value and transmit the peak torque value to be stored in said storage module.

4. The adjustment device as claimed in claim 1, wherein said coupling member of said driving unit includes a sleeve adapted to be sleeved on and drive a polygonal rod of the adjusting unit to rotate, thereby driving an abutment portion of the adjusting unit, which is co-rotatably connected to the polygonal rod, which threadedly engages the shank body, and which abuts against the resilient member, to move along the axis.

5. The adjustment device as claimed in claim 1, wherein said coupling member of said driving unit includes a fastening portion adapted to be fixedly connected to and drive a handle portion of the adjusting unit to rotate, thereby driving an abutment portion of the adjusting unit, which is co-rotatably connected to the handle portion, which threadedly engages the shank body, and which abuts against the resilient member, to move along the axis.

6. The adjustment device as claimed in claim 1, wherein said positioning member includes two clamping portions adapted to clamp the shank body therebetween.

7. The adjustment device as claimed in claim 1, further comprising a calibration unit that includes a calibration member adapted to extend into a calibration hole that is formed in the shank body and that is in spatial communication with the accommodating space, said calibration member being further adapted to be connected to an articulate rod module of the torque wrench, said calibration member being rotatable in a first rotational direction for expanding the articulate rod module to compress the resilient member, and being rotatable in a second rotational direction opposite to the first rotational direction for shrinking the articulate rod module to release the resilient member, thereby varying the preload force of the resilient member.

8. The adjustment device as claimed in claim 7, wherein said calibration unit further includes a calibrating motor disposed on said frame seat, electrically connected to the control unit, and operable for driving rotation of said calibration member.

9. The adjustment device as claimed in claim 8, wherein said calibration unit further includes a restoring member connected between said calibrating member and said calibrating motor for resiliently biasing said calibrating member away from said calibrating motor.
